BRAIN 13

Ketamine

General anesthetic given by injection; a 50:50 mix of two mirror-image forms of the molecule



CLASS
Anesthetic (NMDA receptor blocker)

STUDIED FOR
Anesthesia, depression

HUMAN EVIDENCE
Large trials; FDA approved

US STATUS
FDA approved (Ketalar) · depression off-label

STUDIED FOR

Anesthesia	Depression
HUMAN	HUMAN

Depression use is off-label. The only ketamine-type drug FDA approved for depression is esketamine nasal spray (Spravato), under a REMS.

HOW IT WORKS IN THE BODY

Ketamine ▶ NMDA receptor ▶ Glutamate blocked ▶ Dissociation

- **Blocks the NMDA receptor,** a receptor for glutamate (label)
- **Norketamine,** the main breakdown product, acts there more weakly
- **Dissociative state:** pain relief, with throat reflexes and muscle tone kept
- **Raises** blood pressure, heart rate and cardiac output (label)

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- FDA approved for anesthesia since 1970
- Depression: ketamine vs ECT trial, 403 people
- First depression trial: 7 people, 2000

NOT YET ESTABLISHED

- FDA approval for depression
- Long-term use for depression
- Compounded product quality

QUICK PRACTICAL INFO

HALF-LIFE
10 to 15 minutes (first phase, after injection into a vein)

ROUTES STUDIED
Injection into a vein or a muscle (label)

PHARMACOKINETICS
Liver metabolism (CYP2B6, CYP3A4); active metabolite norketamine

SOLD AS
Prescription injection (Ketalar, generics); Schedule III

MONITORED IN TRIALS
Liver tests with repeated doses (label)

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved, anesthesia (Ketalar, 1970)
FDA	Depression use is off-label
DEA	Schedule III controlled substance
WADA	Not named on the 2026 list

REFERENCES

1. FDA, Ketalar prescribing information (DailyMed), Mar 2026
2. N Engl J Med, Ketamine vs ECT for nonpsychotic treatment-resistant depression, Jun 2023
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

adonismenshealth.com